



Sultan Qaboos University
College of Engineering
Department of Mechanical and Industrial Engineering

MEIE4201 – Ergonomics and Safety Spring 2023

Office ergonomics: Evaluation of Chairs in Office Workstations

Prepared by:

Faisal Alyafaey (125383)
Abdulrahman Almansoori (128460)
Talib Alqarwashi (118731)
Ali Alrawahi (122068)

Instructor:

Mohammed Othman

Abstract

Ergonomics, often known as human factors engineering (HFE), is an approach that tries to design environments or workplaces that are user-friendly for humans. Ergonomics aims to increase efficiency while reducing fatigue. An uncomfortable workplace can quickly wear people out and result in lower performance, making the environment an essential factor in deciding productivity and efficiency. The field of ergonomics assists in gaining an understanding of human requirements and constraints for certain jobs, enabling the development of products and procedures that enhance people's interactions with the products and the surroundings.

The risk of injury, mistakes, and accidents can be reduced by putting ergonomics concepts into practice. Designing for humans is the focus of ergonomics, which in an office setting applies to many kinds of objects like, desks, chairs, stands, lighting, monitor, and other requirements for a workplace. The importance and advantages of office ergonomics are shown in this report's investigation. The report will also include an outline for office ergonomics and discuss the various difficulties and injuries that might result from inadequate workplace ergonomics.

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1. Introduction

Designing workstations and environments that are suitable and comfortable for people to use is the focus of ergonomics, often known as human factors engineering. It is the science of customizing workplaces, tools, and layouts to meet the needs of the user. Ergonomics' main objective is to increase productivity and efficiency by reducing human discomfort and stress.

It tries to create systems, workstations, and products that suit people's needs while taking into consideration their physical, cognitive, and emotional capabilities and limits. This could involve altering the dimensions, designs, and placements of tools and machines. Furthermore, enhancing the lighting, ventilation, and other aspects of the environment can also help.

Ergonomics can boost performance and productivity: When employees are comfortable, and their physical and mental workloads are optimized, they are more likely to perform well and be productive. Moreover, Ergonomics can enhance product design by helping to create products that are more user-friendly and efficient by considering user needs and abilities.

It's critical to understand ergonomics and how it relates to the workplace if you want to prevent industrial accidents. This involves evaluating elements like the height of your workstation and the angle of your computer monitor in addition to considering how your body feels after a long day at work. Ergonomics helps reduce strain from repetitive actions and discomforts like tired eyes or painful wrists by adapting equipment to the user and encouraging proper posture.

For example, Office ergonomics involves redesigning or evaluating office chairs. Continuous sitting in an uncomfortable chair can cause pain, discomfort, and even serious medical conditions like musculoskeletal illnesses. It is essential to design or evaluate workplace chairs that offer the right support and encourage healthy posture. The seat should support the person's legs, and the backrest should support the lower and upper backs. Furthermore, for the person's arms and shoulders to receive comfort and adequate support, armrests should be flexible.

2. LITERATURE REVIEW

This literature review attempts to look into studies done on the evaluation or redesign of office chairs to improve ergonomics.

First, research by Amick III and colleagues [1] from the research paper “Dissemination and use of a participatory ergonomics guide for workplaces” The study's goal was to evaluate the effectiveness of an ergonomic chair with lumbar support and seat depth adjustments. The use of ergonomic seats, according to the study's findings, significantly reduced back pain and suffering while boosting employee productivity.

Second, a study was done by Robertson and colleagues [2] from the article “Improvements in musculoskeletal health and computing behaviors: Effects of a macro ergonomics office workplace and training intervention.” The article was directed to look at how efficient a seat with a movable seat and backrest is in reducing outer muscle problems and upgrading worker productivity. The investigation uncovered that the utilization of an ergonomic seat enormously limited the possibilities of outer muscle issues and furthermore helped the fulfillment of the workers.

A third study was conducted by Hedge et al. [3] by the name “Handbook of human factors and Ergonomics methods” The study found that back pain was reduced and comfort was improved in office chairs that come with adjustment options of lumbar support and seat depth.

The fourth study by the name of “ergonomics in Dentistry: An Ounce of Prevention is Better than Pounds of Cure,” was done by Kumar et al.[4], at the end of the study, it was found that ergonomic office chairs reduced back pain and improved spinal posture.

3. Methodology and Recommendations

The significance of office ergonomics in maintaining your health cannot be highlighted. Sick days are frequently taken off due to work-related injuries, especially repetitive strain injuries. It's crucial to adjust your workstation to suit your unique needs. In many cases, poor office ergonomics led to musculoskeletal problems and eye issues. Some main issues that arise from poor ergonomic designs are:

- a) **Back and Neck Pain:** These are usually the most common issues of office workers. Degenerative disc problems or excessive strain on the spine are two potential causes of this. As the spine and other bones shift position over time because of poor posture, back and neck pain occurs.
- b) **Repetitive Strain Injuries:** Repetitive strain injury can be caused by sitting in an unpleasant, unnatural position on a regular basis. It refers to muscular, nerve, and tendon pain caused by overuse and repetitive movements. While it is most associated with typing-related wrist and hand injuries, it can also refer to several painful disorders affecting the muscles, tendons, nerves, and other soft tissues. Most repetitive strain injuries are caused by nerves in the upper body, from the neck and shoulders down to the wrists and hands.
- c) **Vision and Headache Issues:** The term "computer vision syndrome" refers to a group of eye and vision conditions brought on by extended computer use. People who spend a lot of time at a computer frequently experience headaches. They are commonly brought on by too much screen glare as well as by poor computer brightness and color.

There are many factors that should be considered when applying an ergonomic assessment to an office. Chairs, tables, monitors, keyboards, mice, Lighting, temperature, vibration, posture, and noise are all factors that can contribute to getting a better office design.

The team established the following guidelines to assess the design of an office chair.

1. Check the chair's backrest to see if it maintains the natural curve of the spine and provides proper lumbar support. Look for a seat pan that is wide enough and with a contoured border that does not crush the worker's legs at the rear.
2. Determine whether the chair has features such as adjustable seat height, backrest height, seat pan tilt, and armrest height. These elements should be simple to modify and offer a wide range of adjustments to accommodate the worker's height and body size.
3. Sitting posture: While sitting in the chair, observe the worker to see if their feet are flat on the floor, their hips are slightly higher than their knees, and the backrest supports their back. Check to see if the armrests support their elbows comfortably.
4. Ask the worker if they are uncomfortable or in pain when sitting in the chair.
5. Make any other chair adjustments required to get a comfortable and supported posture.



Figure 1: chair example 1



Figure 2: chair example 2

Figure 1 shows a bad design of a chair, and figure 2 shows a well-designed chair using the guidelines the team established earlier.

Moreover, sitting posture is a very important aspect that should be looked upon. If the posture is bad, then even if the design is good, there will be injuries in the long term. Figure 3 shows how a good posture can be obtained. While figure 4 shows examples of good and bad postures.

1. Maintain your feet flat on the floor to maintain your backbone balance and make it easier to bend your lower back.
2. Keep your elbows to the side of your body to maintain an L-shape at the elbow joint.
3. Sit up straight and keep your back straight to allow your bones, tendons, muscles, and joints to align.
4. Make sure you're sitting with your hips far back in your chair.
5. Ideally, your knees and hips should be in line.



Figure 3: Good posture



Figure 4: Bad postures

There are many recommendations that can improve the overall comfort of workers in an office when thinking of chairs. Here are a few:

1. Armrests: Armrests should be adjustable and designed to support the user's shoulders and arms to enable relaxed and pleasant mouse and keyboard use.
2. Work surface height: The ability of the user to easily adjust the desktop, which is commensurate with the length of the arm and wrist while writing or using the laptop.

3. Footrest: Footrest accessories are important for the user during work, as they are a footrest and also protect the lower back due to pressure as shown in figure 5.



Figure 5: Footrests

4. Seat material. In order to be comfortable to sit in for prolonged periods of time, the office chair's seat and back material should have sufficient padding, typically constructed of breathable materials that reduce the development of heat and moisture. Some popular materials for office chair seats include:
 - Mesh: This material is more common in seats because of its lightweight and allows air to pass through.
 - Fabric: There are many seats that contain fabric, and they differ according to the color and fabrics used in them. One of its most important features is that it is more comfortable and allows air to pass through
 - Leather: This material is used in luxury seats; it is very comfortable and durable
5. Seat Width: The seat width varies according to the user and ensures the user's comfort in the seat. Therefore, we found that the average seat width ranges from 16 to 18 inches.
6. Stability: It is preferable to use a seat with a five-point base to ensure the stability of the seat and ease of movement, as shown in Figure 6.



Figure 6: Good base chair

7. Swivel: Any traditional or ergonomic chair should be simple to rotate so the user can easily reach different parts of their desk without straining.

TABLE 9.5 Recommended Dimensions for Office Chairs (Dimensions in Centimeters, Angles in Degrees)

Feature	BS 3079 and 3893 ^a	CEN ^b	Diffrient et al. (1994)	Danero & Zelnik	Grandjean (1980)	DIN ^c	SS ^d
Seat							
Height	43–51	39–54	35–52	36–51	38–53	42–54	39–51
Width (breadth)	41	40–	41	43–48	40–45	40–45	42–
Length (depth)	36–47	38–47	33–41	39–41	38–42	38–42	38–43
Slope angle	0–5	0–5	0–5	0–5	4–6	0–4	0–4
Backrest							
Top height	33–	—	—	—	48–50	32–	—
Bottom height	20–	—	—	—	—	—	—
Center height	—	17–26	23–25	19–25	30	17–23	17–22
Height	—	10–	15–23	10–20	—	22–	22–
Width (breadth)	30–36	36–40	33	25	32–36	36–40	36–40
Horizontal radius	31–46	min. 40	31–46	—	40–50	40–70	40–60
Vertical radius	convex	—	—	—	—	70–140	convex
Backrest–seat angle	95–105	—	35–100	95–105	—	—	—
Armrest							
Length	22	20–	15–21	—	—	20–28	20–
Width (breadth)	4	4–	6–9	—	—	—	4–
Height	16–23	21–25	18–25	20–25	—	21–25	21–25
Interarmrest width	47–56	46–50	48–56	46–51	—	48–50	46–
Feet number	—	—	—	—	5	—	—
Base number	—	—	—	—	40–45	—	—

^aBritish standards.

^bEuropean standards.

^cGerman standards.

^dSwedish standards.

The table provided above displays the recommended dimensions for office chairs. When designing office chairs, it is imperative to take these dimensions into account to ensure that the produced chairs are ergonomically well-designed.

Remember, As there is no one-size-fits-all approach to ergonomics, it is crucial to take into account the particular requirements of each person and make adjustments as necessary.

Other recommendations to increase productivity in an office:

1. Putting your face too close to a screen might cause eye strain. Neck injuries are common when the display is positioned at an unusual angle or too far away from your eye level. Your computer monitor's top should be level with your eyes.
2. Adjust the brightness of your monitor to match the brightness of the surrounding environment.
3. Reduce how much time you spend on the phone. If you spend a lot of time on the phone at work, consider replacing your cell with a headset. If you hold your phone between your ear and shoulder, the muscles in your neck can become strained.
4. Crossing your legs may result in health problems owing to bad posture. When your legs are crossed, the peroneal nerve, which provides sensation to your legs and feet, is compressed.
5. Check that your forearms are straight and parallel to the floor of your chair. As a result, repetitive strain injuries are prevented.
6. Taking frequent short breaks is better for your back than taking fewer longer ones.

These are just a few suggestions for office workstation ergonomics evaluation. By putting these recommendations into practice, employees' comfort, productivity, and general well-being can all be enhanced.

4. Discussion

The methodology outlined in this report is significant for anyone who spends extended periods sitting at a desk. It emphasizes the critical role of office ergonomics in maintaining one's health and the adverse effects of poor ergonomic design on musculoskeletal problems and eye issues. The report provides a comprehensive guide for assessing the design of an office chair, which is a crucial component of office ergonomics, and recommends guidelines for individuals to ensure their chair provides necessary support and adjustability.

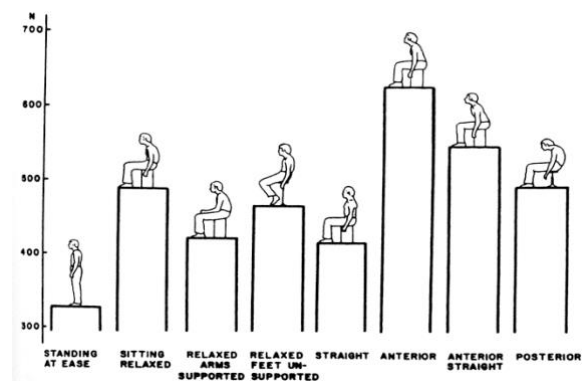


Figure 7: Bad sitting postures.

The image depicted in Figure 1 above illustrates a poorly designed chair that lacks proper back support. Chairs of this type can cause chronic pain for their users due to their lack of ergonomic design. In contrast, the chair depicted in the other figure 2 serves as a prime example of an ergonomically well-designed chair. This chair features back support and elbow holders, both of which are adjustable in height to provide maximum comfort for the user.

Figure 7 depicts various sitting postures and the corresponding levels of stress exerted on the spine. When designing workplace chairs, it is crucial to consider such studies. Among the depicted postures, the "Anterior" sitting position is the most detrimental to the spine, while the "Straight" posture is the most beneficial for the human back.

Figure 8 provides a detailed explanation of proper sitting posture while working at a desk in an office. It demonstrates the best way to sit without causing damage to the user's back. As shown, lifting weight while sitting places the highest stress on the back, whereas sitting in a relaxed position is best for the back and reduces strain.

Overall, figure 3 serves as a useful guide for maintaining proper posture while seated, which is essential for preventing back pain and discomfort associated with long hours of desk work.

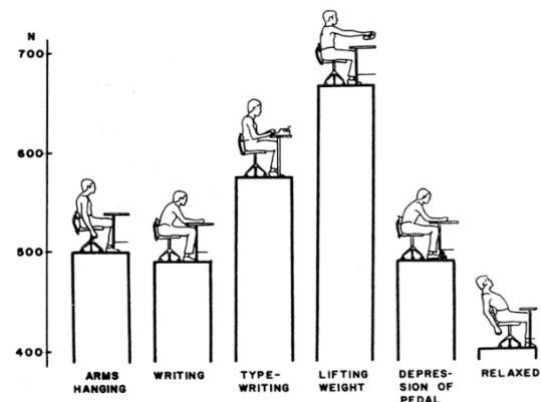


Figure 8: Good sitting postures

5. Conclusion

In summary, this paper emphasizes the value of workplace ergonomics and offers a detailed manual for maintaining effective office chair ergonomics. The probability of developing musculoskeletal issues, vision problems, or other health problems can be decreased by employers and employees working together to ensure that the workplace environment is modified to meet each person's specific needs.

6. Reference:

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